

AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Project Description

The AI-Powered Workforce Analytics & Talent Intelligence Dashboard is an intelligent HR analytics platform that transforms workforce data into actionable business insights. The solution integrates SAP SuccessFactors, Workday, Oracle HCM, Learning Management Systems (LMS), recruitment portals, employee performance reviews, engagement surveys, and attrition history. Using Machine Learning (ML), Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG), the platform predicts attrition risks, identifies skill gaps, analyzes workforce health, monitors diversity metrics, and provides AI-powered recommendations through interactive dashboards and conversational analytics.

Data Sources

SAP SuccessFactors • Workday • Oracle HCM • Recruitment Portals • LMS • Performance Reviews • Engagement Surveys • Attrition History

Models Suitable for Implementation

Model	Use Case	Expected Performance
Logistic Regression	Attrition Prediction	84–89%
Decision Tree	Workforce Classification	86–90%
Random Forest	Talent Risk Prediction	92–96%
XGBoost	Attrition & Workforce Analytics	95–98%
LightGBM	Workforce Forecasting	94–97%
CatBoost	Categorical HR Data	94–97%
ANN	Pattern Learning	91–95%
BERT	Employee Feedback Analysis	F1: 92–95%
Llama 3 + RAG	Conversational HR Intelligence	LLM-based

Recommended Solution

XGBoost is the most suitable model because workforce data is predominantly structured and requires accurate prediction of attrition, workforce trends, and talent risks. **Random Forest** and **LightGBM** are strong alternatives for predictive analytics, while **BERT** enhances employee feedback analysis. **Llama 3 integrated with RAG** enables conversational workforce intelligence, executive reporting, and natural language querying.

Expected Outcomes

- Workforce Performance Analytics • Attrition Risk Prediction • Skill Gap Identification • Diversity & Inclusion Monitoring
- AI-Powered Workforce Planning • Conversational HR Analytics